ZHAN KANG NG

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EDUCATION

Singapore University of Technology and Design (SUTD) **Sept 2024 – July 2028 (Expected)**
B.Eng., Computer Science and Design (Artificial Intelligence)

Singapore Polytechnic **Apr 2019 – Apr 2022**
Diploma in Information Technology

EXPERIENCE

Undergraduate Research Assistant (Decentralized Federated Learning) | GitHub Repo **Oct 2025 – Oct 2026**
SUTD – Supervised by Dr. Marie Siew

- Converted an open-source federated learning simulator into a live peer-to-peer system across 3 NVIDIA Jetson Orin Nano devices, to test decentralized learning on real edge hardware instead of in simulation
- Built the gRPC and Protobuf transport layer for asynchronous PyTorch model exchange, plus zero-config peer discovery that scans about 16,000 addresses in 25 seconds
- Extended peer selection with an Earth Mover's Distance term for data skew, and added IID, Dirichlet and shard splits
- Packaged the system as a Docker image on GHCR so any Jetson joins the network with one command

Software Engineer Intern (Machine Vision) **Sept 2021 – Feb 2022**
Inspiraz Technology Pte. Ltd.

- Co-developed a screw inspection tool for the WizerPro platform in C++ (Halcon), measuring pitch, angle and diameters
- Automated thread direction and screw region detection, removing manual operator selection
- Diagnosed why the legacy algorithm failed on larger screws and prototyped best-fit line and intersection methods to locate thread peaks
- Debugged DLL dependency and crash issues, and wrote the developer documentation

PROJECTS

Multiplayer Terminal Tetris (Systems Programming, C) | GitHub Repo **Jul 2026 – Aug 2026**

- Built the admin control channel for an epoll-based multiplayer Tetris server: a Unix socket and its CLI client
- Added the server's HTTP-style protocol layer, and wrote 101 unit tests for its message parser (team of 3)

AttendEase (Android NFC attendance app, Java) | GitHub Repo **Mar 2026 – Apr 2026**

- Built an Android app in Java (MVVM, Room) where lecturers take attendance by tapping students' NFC cards

Memory Game on FPGA (Computation Structures, Lucid) **Mar 2026 – Apr 2026**

- Built a Simon-style memory game on an FPGA as a microprogrammed CPU: an FSM driving a 32-bit ALU datapath
- Co-wrote the control FSM, and wrote the integration and unit testbenches and module docs (team of 4)

HealthyHelper Bot (SIM-UOL CSSC Hackathon) | GitHub Repo **Nov 2025**

- Co-built a Telegram diet assistant on the Perplexity API (RAG pipeline for meal plans); placed 4th of 16 teams

Machine Learning Capstone Project (United Overseas Bank) | GitHub Repo **Apr 2021 – Aug 2021**

- Cleaned 11,000+ scraped company webpages, handling stray characters and mixed Southeast Asian language text
- Benchmarked BERT, VDCNN and MLP multi-label classifiers on the cleaned data across 4 business categories
- Led the team's client meetings for part of the project, and delivered the working tool to the bank at handover

SKILLS

- **Languages:** Python, TypeScript, Java, C, C++, JavaScript, HTML/CSS, SQL, Bash
- **AI & ML:** PyTorch, Transformers (BERT), RAG Pipelines, NLP, Computer Vision (Halcon), LLM APIs
- **Infrastructure & Tools:** Docker, gRPC/Protobuf, NVIDIA Jetson, Git/GitHub Actions (CI/CD), Linux